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ROSABIO

Decoding Genetic Complexity to Accelerate Clinical Breakthroughs

Non-confidential presentation
March 2026

Old Paradigm, New Reality

Old Paradigm:

- Reference genome-to-phenotype associations are based on a paradigm developed in a different era: assembly and alignment to a reference genome
- Effective but has inherent limitations and costs that limit clinical applications

New Reality:

- NGS technologies now generate massive amounts of genomic data annually
- Increasing importance of precision medicine and genetic screening
- Advances in ML and AI can allow new insights

Improved, more accurate sequence analysis would enable new data insights

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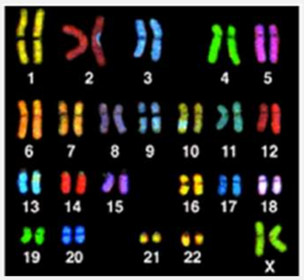
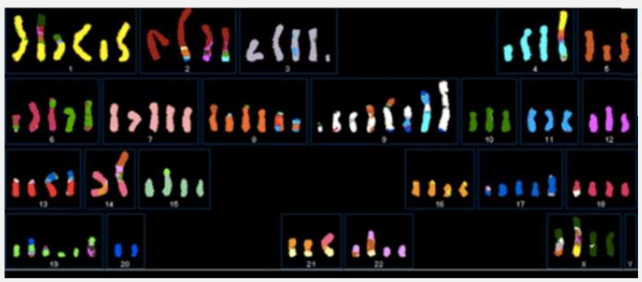
Reference Genomes Poorly Represent Biology

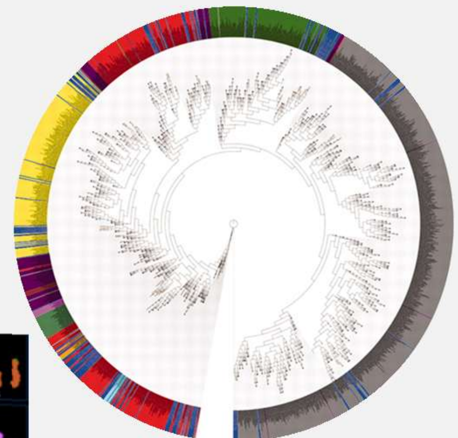
Normal (reference) vs. Cancer





https://bscb.org/learning-resources/softcell-e-learning/what-causes-cancer-more-details/

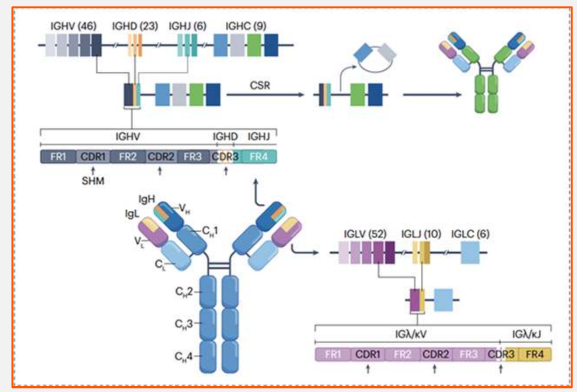
Human Microbiome



DOI: 10.1126/science.1183605

Immunology

Adaptive Immune Receptor Repertoire



https://doi.org/10.1038/s43586-023-00284-1

YET, current methods are dependent on comparisons to reference genomes

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Existing Genomics Workflows Involve High Complexity, Substantial Time And Resource Requirements, And Unavoidable Data Loss

Primary Analysis

Genomic DNA → WGS / NGS on gene panel

fastq

- On-instrument step and requires many workflows need deep NGS to see signal - $$$
- Upfront loss of data when using gene panels

Secondary Analysis

Batch correction & normalization → Genome or pan-genome alignment, Assembly

- Computationally intensive - $$$
- Many hours to run
- Degree of data loss depends on the reference

Tertiary Analysis

Extract features
- SNV
- CNV
- Indels
- Fusions, VDJ

→ Downstream statistical analysis, classification, variant reporting, ML, AI, etc.

Numerous steps – all with data loss- are required prior to AI

Results in need for a separate analysis workflow for each sample type – RNA, single cell—adds additional complexity, resources and time

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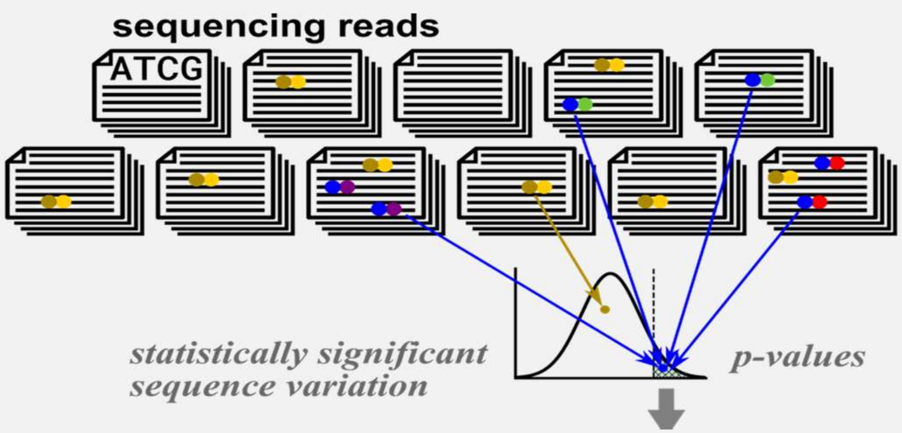
ROSA BIO Changes The Paradigm With A “Statistics First” Approach

- Completely reference-free – uses all the data, eliminating the need for alignment to a reference genome
- Robust statistical analysis for upfront extraction of signal from noise to generate AI-ready data.
- Can find signals that are missed by reference methods.

Proof of principle for statistics-first in SPLASH:
A statistical, reference-free genomic algorithm unifies biological discovery



Ref: Cell Volume 186, Issue 25 P5440-5456.

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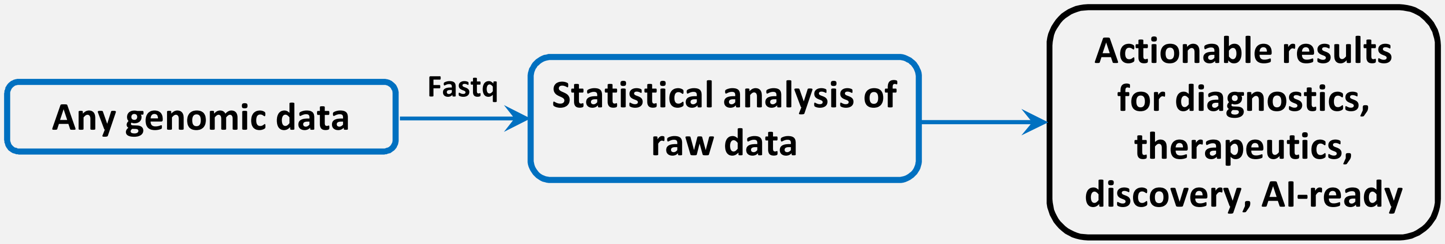
ROSA BIO's Sequence Analysis Platform

- **PANOPTIC**: Detects all sequence variation, epigenetic and other modifications in raw nucleic acid sequence data
- **FASTER**: >10X current analysis workflows
- **CHEAPER**: Computationally light and scalable



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ROSA BIO's Alternative Approach: Statistics-first Analysis Of Raw Sequence Data

PUBLIC DOMAIN

ALIGNMENT WITH REFERENCE

SPLASH

FEATURIZERS

Reference Alignment-based

Variant calling (e.g.,: DRAGEN)
Targeted panel-based MRD

ROSA BIO

ROSA-V variant detection

Structure-based MRD

P2G

ROSA BIO Platform: Advantages

Technical Advantages:

- Statistics-first approach detects sequence complexity and variation missed by other methods
- Compatible with any analyte (e.g., tissue sample or plasma, cfDNA, RNA)
  - single cell or bulk
- Does not require application-specific tuning

Scientific Advantages:

- Consistent high-performance; agnostic to nucleic acid type: DNA (WGS and targeted), RNA, short and long read, exome, non-coding regions, 5-methylcytosine methylation, 5-hydroxymethyl cytosine, bisulfite, etc
- Detects structure-based changes, not based on targeted panels

Operational Advantages:

- Panoptic , faster, cheaper nucleic acid sequence analysis
- Readily scalable for diverse applications: Dx, Tx, translational research
- Can analyze terabytes of data in ~1 day
- Output = AI-ready

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ROSA BIO Proprietary Bioinformatics Tool Functionalities

- **COMB:** calling engine: uses full-read lengths, including long read sequencing data
- **Rosa-V**: variant sequence caller: superior detection of variants not identified with alignment methods, especially in regions with repeat sequences (ca. 50% of human genome)
- **Syve**: disease vs non-disease classifier
- **P2G**: can predict a genotype from analysis of nucleic acid sequences derived from a source with the phenotype of interest
- Others in progress

All data generated is AI-ready and suitable for integration into multi-omics analysis

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ROSA BIO Therapeutic Applications

Patient Stratification

- ID patient subpopulations with differential response to therapy
- Selection of optimal drug therapy for patients
- Selection of patients for clinical trial enrollment

Drug Discovery

- Variant calling
- Mutations in non-coding regions
- Phenotype (e.g., resistance) gene-directed genome mining
- MOA studies

Drug optimization

- Antibody/ADC engineering
- Characterization of binding site interactions
- Optimization of protein-protein interfaces

Applicable Modalities

- Small molecules
- Proteins including Ab/ADC
- RNA drugs
- Gene therapy

Pharmacogenomics

- Toxicology profiling of candidates
- DILI

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TX Example 1: Detection of Sequences Encoding Aromatic Polyketide Synthases

Data derived from to naphthalenone biosynthetic pathway in fungi

- cDNA sequences of functional and non-functional PKS orthologs input into Rosa P2G
- P2G signature of conserved amino acid sequence – bypassing multiple sequence alignment

P2G re-identified the functional PKS *ab initio* in *Aspergillus parvulus* and other species

- Constitutes a general strategy for identify BGCs in fungal genomes
- This and other approaches provide a general strategy for discovery significantly expanding current phenotype, e.g., resistance-gene encoded, searches

Tx Example 2: Patient Stratification

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- Many approved drugs only work patient subpopulations or have side effects in subpopulations
  - Non responder examples;
    - ICIs: 20-40% benefit
      - PD-L1 inhibitors: 60-80% of solid tumors do not have a durable long term response
    - GLP-1: 10-20% of patients are non-responders
    - Epilepsy: 30-40% of patients have drug resistant epilepsy
    - Statins: 10% of patients are non-responders
  - Selection of treatment based on patient genotype can provide improved outcomes and reduced costs
    - Examples:
      - MATADOR study gene profile for taxanes vs alternative therapy for breast caner, iScience 27,110425 (2024)
      - Response to platinum and PARP inhibitors in BRCA associated pancreatic cancer *Cancer Discov* (2023) 13 (8): 1826–1843
      - Cancer agnostic, mutation-specific drugs: pembrolizumab, larotrectinib, dabrafenib

- Drug development efficacy can be improved with selection of an appropriate patient population

Rosa tools can identify patient subpopulations based on pre-treatment ctDNA blood sample

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ROSA BIO Dx Applications

Platform with Broad Utility

- Faster, better cheaper analysis of NGS data
- Results orthogonal to alignment-based approaches

Disease identification, classification and monitoring

- Early detection: cancer and other conditions
- Tumor profiling
- Disease subtype classification
- Measureable residual disease (MRD)

Especially powerful for diseases characterized by:

- Genetic complexity
- Alternative Dx methods invasive, unpleasant, insensitive
- Unmet need

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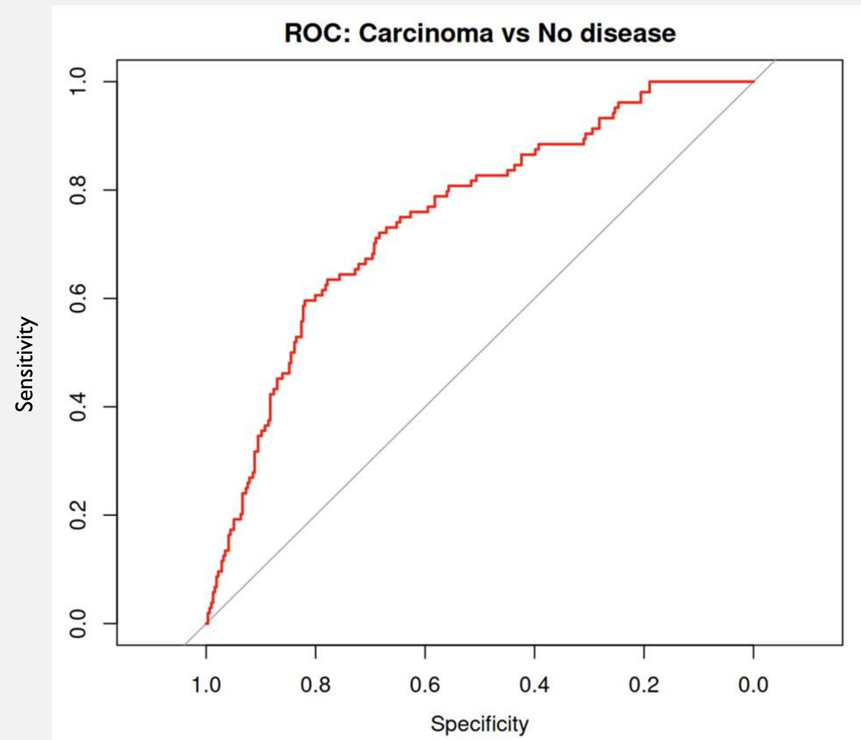
Dx Example: Breast cancer detection with targeted bisulfite methylation sequencing

- Empirical subset of data for training
- Classifier developed based on signature
- Normals and cases defined:
    - NORMAL= fibrocystic changes; No disease, atypical hyperplasia, fibroadenoma
    - CASES: huminal A/B (+/-Her2); I-III
- Syve AUC .75 for Stage I and II
    - no published AUC



Projected Trajectory

Initial focus on collaborations with commercial partners

- Validation
- Expand access to datasets of interest
- Sell/license initial assets to advance company
- Avoid fee for service model

Collaborate with companies with complementary assets

- Sequence data and biological or clinical metadata
- Domain specific assets
- Tx, Dx, agriculture
- Share in product revenue

Integration sequence data other data modality(ies)

- Mass spec, image, clinical data, etc
- ML and AI

Foundation model

ROSA BIO: Strong intellectual property

- ROSA statistics-first approach is differentiated
- Foundational patent applications filed; solely owned by Rosa
- Proprietary platform software; solely owned by Rosa Bio

ABOUT ROSA BIO

LEADERSHIP TEAM

Julia Salzman, PhD
Founder

Associate Professor of

Biomedical Data Science & Biochemistry and by Courtesy, Statistics, Stanford University

- Inventor of SPLASH (Cell, PNAS, Nature Biotechnology)
- Sloan Research Fellow
- NSF CAREER Awardee

Michael Rabson, PhD
CEO

Formerly:

- Partner at WSGR
- General Counsel at Maxygen and Cytokinetics